

L12: Generating Networks with Community Structure

In the previous models, our focus has been on the degree distribution. That is an important property of a network – but it is not the only key feature.

As we saw earlier, another common property of many real-world networks is clustering and the presence of communities.

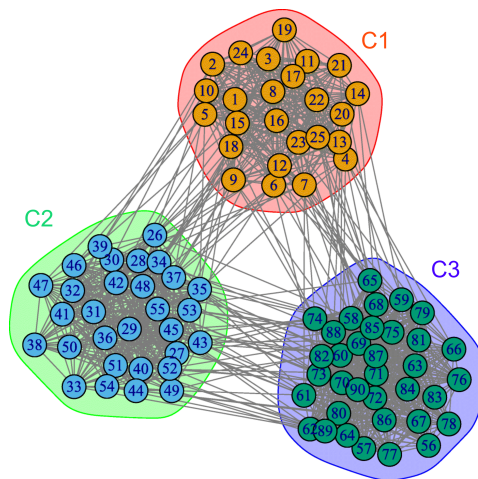
There is no reason to expect that the PA model and its many variants can generate any non-trivial community structure.

In fact we have already seen some models earlier that can generate clustering and/or communities. Recall the *Watts-Strogatz* model from Lesson-5: it can generate strong clustering (similar to that in regular mesh networks) – but it cannot generate communities and its degree distribution is not a power-law.

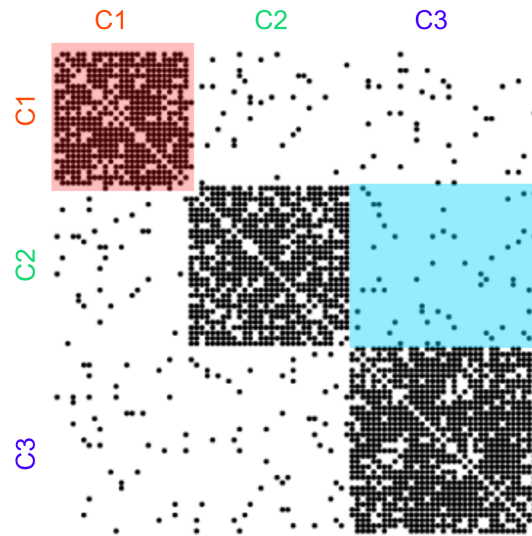
Additionally, in Lesson-8 we presented two methods to generate networks with a given community structure: the *Girvan-Newman (GN)* model and the LFR method. Recall that the latter can generate networks with both a power-law degree distribution and a power-law community size distribution.

Here, we briefly present one more model to construct networks with strong clustering and community structure: the **Stochastic Block Model (SBM)**. The model description is very simple: we are given the number of nodes n , the number of edges m , the number r of communities and their sizes $n_1, n_2, \dots, n_r$ (with $\sum_{i=1}^r n_i = n$), and a symmetric r -by- r probability matrix P .

The (i,j) element of P is the probability that a node of community i connects with a node of community j , i.e., $P(u \longleftrightarrow v \mid u \in i, v \in j)$. The diagonal elements of P represent the probability that nodes of the same community connect to each other – those elements are typically larger than the non-diagonal elements. The rows and columns of P are not probability distributions, hence they do not necessarily sum to one.



Adjacency matrix (as a dot matrix)



Intercommunity Connection Density Matrix

$$\begin{matrix}
 & \begin{matrix} C1 & C2 & C3 \end{matrix} \\
 \begin{matrix} C1 \\ C2 \\ C3 \end{matrix} & \begin{bmatrix} 0.8 & 0.05 & 0.05 \\ 0.05 & 0.8 & 0.05 \\ 0.05 & 0.05 & 0.8 \end{bmatrix}
 \end{matrix} = P$$

The adjacency matrix is represented in dot format above (as opposed to the usual numerical format). Each entry is either blank if there is no edge between respective nodes and filled with a dot if there is. In relation to the three communities shown above, the adjacency matrix can be thought of as being broken into 9 blocks, one block representing the pairings between community 1 and itself, another block representing the pairings between community 1 and community 2, and so on. Each block in the adjacency matrix is distilled into a single number in P . For example, the red-highlighted block represents all pairings (u, v) where u and v are both in community 1. About 80% of the possible entries in this block are filled in with a dot, so the value in P (also highlighted in red), is 0.8. In another example, the cyan-highlighted block represents pairings (u, v) where u is a node in community 2 and v is a node in community 3. About 5% of these entries are filled in, so the corresponding entry in P is 0.05.

The visualization shows a network with $r=3$ communities and $n=90$ nodes ($n_1 = 25, n_2 = 30, n_3 = 35$) generated by a SBM model. The visualization also includes the 3-by-3 probability matrix P and the 90-by-90 adjacency matrix of the resulting network.

Note that SBM models can generate a clearly defined community structure and strong clustering within each community – but it does not control the degree distribution or other network properties. Additionally, the communities are “flat” – without any hierarchical structure.

There is a rich literature, mostly in statistics and machine learning, that focuses on the estimation of the SBM model parameters from network data.

Later in this lesson we will study a more general approach, referred to as **Hierarchical Random Graph** (HRG) model, and we will present there a statistical approach to estimate the model parameters.

Food For Thought

Identify the similarities and differences between the SBM, GN, and LFR models in terms of their network properties.